

Project Report: House price prediction

NAME: DEVANAND.R

REG NUMBER: 25BHI10067

COURSE: BTECH HEALTH INFORMATICS

1. Introduction

House price prediction is the process of estimating the market value of a property by analysing its core components. While location is a primary driver, the internal configuration—specifically the **Bedroom-Bathroom-Kitchen (BBK)** ratio—serves as the most significant indicator of a home's utility and target demographic.

2. Objective

The goal of this project is to analyse how specific configurations of living spaces (Bedrooms, Bathrooms, and Kitchens) correlate with price fluctuations in the residential market.

3. Key Feature Analysis

A. Bedrooms: The Capacity Indicator

The number of bedrooms is the primary determinant of a house's "size category" (e.g., 2BHK, 3BHK).

- **Master Bedrooms:** Usually command a 15–20% premium if they include ensuite bathrooms and walk-in closets.
- **Guest/Study Rooms:** Added versatility that increases the resale value for work-from-home professionals.

- **Impact on Price:** Generally, every additional bedroom increases the property value by **10% to 15%**, assuming the total square footage increases proportionally.

B. Bathrooms: The Luxury & Convenience Factor

Modern real estate trends show that the bathroom-to-bedroom ratio is increasingly critical.

- **Full vs. Half Baths:** A "Full Bath" (Toilet, Sink, Shower/Tub) adds significantly more value than a "Half Bath" (Powder room).
- **Fixture Quality:** High-end tiling, water-saving fixtures, and ventilation systems are secondary price drivers.
- **The Ratio Rule:** Houses with a 1:1 bedroom-to-bathroom ratio typically sell faster and at higher price points than those with shared facilities.

C. Kitchens: The Heart of the Home

The kitchen is often cited by appraisers as the room that "sells the house."

- **Layout Types:** Open-plan kitchens (connecting to the living area) are currently valued higher than closed, traditional layouts.
- **Functionality:** Features like a central island, pantry space, and built-in appliance "garages" drastically increase the "utility value."
- **Materials:** Granite or quartz countertops and modular cabinetry can increase a kitchen's contribution to the total house price by up to **8%**.

4. Comparative Valuation Table

The following table illustrates how variations in these three features typically affect the pricing of a standard urban apartment.

Feature Configuration	Target Audience	Price Impact
1 Bed / 1 Bath / Standard Kitchen	Single Professionals	Base Price
2 Bed / 2 Bath / Modular Kitchen	Small Families	+35% to 45%
3 Bed / 3 Bath / Gourmet Kitchen	Large Families	+70% to 90%
4+ Bed / 4+ Bath / Dual Kitchens	Luxury/High-Net-Worth	+150% and above

5. Functional Requirements for Prediction

To predict a price based on these features without complex software, we use **Comparative Market Analysis (CMA)**:

1. **Selection of Comparables:** Identify 5-10 houses in the same radius.
2. **Feature Adjustment:** If the "Subject House" has an extra bathroom compared to the "Base House," a standard "Value Add" (e.g., \$5,000–\$10,000) is added to the estimate.
3. **Depreciation Factor:** Subtract value based on the age of the kitchen cabinets or bathroom plumbing.

6. Conclusion

Predicting house prices through the lens of internal features reveals that buyers prioritize **privacy (Bathrooms)**, **rest (Bedrooms)**, and **utility (Kitchen)**. A house with an optimized BBK ratio will consistently outperform houses with larger total square footage but inefficient room distributions.